

MIT SLOAN SCHOOL OF MANAGEMENT
MIT COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE LABORATORY (CSAIL)



MODULE 6 UNIT 2
Assignment

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Learning outcomes:

LO3: Propose an initiative for a specific business application of AI.

LO4: Produce a roadmap for an organization to gain strategic advantage using AI.

Plagiarism declaration

1. I know that plagiarism is wrong. Plagiarism is to use another's work and pretend that it is one's own.
2. This assignment is my own work.
3. I have not allowed, and will not allow, anyone to copy my work with the intention of passing it off as their own work.
4. I acknowledge that copying someone else's assignment (or part of it) is wrong and declare that my assignments are my own work.

Name: Karthekeyan Anumanpalli Kuppuraj

1. Instructions and guidelines (Read carefully)

Instructions

1. Insert your name and surname in the space provided above, as well as in the **file name**. Save the file as: **First Name Last Name M6U2 Assignment** – e.g., **Lilly Smith M6U2 Assignment**. **NB:** Please ensure that you use the name that appears in your participant profile on the Online Campus.
2. Write all your answers in this document. There is an instruction that says, "Start writing here" under each question. Please type your answer there.
3. Submit your assignment as a **Microsoft Word document only**. No other file types will be accepted.
4. Do **not delete the plagiarism declaration** or the **assignment instructions and guidelines**. They must remain in your assignment when you submit.

PLEASE NOTE: Plagiarism cases will be investigated in line with the Terms and Conditions for Participants.

IMPORTANT NOTICE: Please ensure that you have checked the Online Campus for the due date for this assignment.

1. There are **6 pages** and **1 question** in this assignment.
2. Make sure that you have carefully read and fully understood the questions before answering them. Answer the questions fully but concisely and as directly as possible. Follow all specific instructions for individual questions (e.g., “list,” “in point form”).
3. Answer all questions in your own words. Do not copy any text from the notes, readings, or other sources. **The assignment must be your own work only.**

2. Question

For each assignment submission included in Modules 1 to 5 of this program, you were asked to complete activities that should now help you complete a final, polished AI roadmap. The roadmap will function as a starting point for the successful deployment of AI in an organization of your choice, with the aim of achieving competitive advantage.

For this final assignment, use your answers from previous modules as a guide to populate the roadmap template. You should use the organization that formed the basis of your assignment submissions for Modules 1 to 5 to complete your roadmap. Furthermore, you can use the feedback you have received in previous modules to refine your answers in your roadmap.

Note:

Please ensure that you consult the assignment guide in Unit 2 of this module for details on the specific content required for each section, and **word limit adherence for each section.**

Question 1

Complete the four sections of the following roadmap template.

Executive summary	Summarize your plan for the strategic implementation of AI into your chosen organization. (Max. 500 words.) Start writing here: Our organization operates in a highly regulated financial-services environment where analysts play a crucial role in reviewing customer cases, gathering data from numerous systems, and writing detailed investigative narratives. Currently, analysts spend nearly 70% of their time on manual data collection, validation, and synthesis before they can begin interpreting findings. This labor-intensive approach introduces variability, increases turnaround times, and contributes significantly to operational costs. As volume grows and regulatory expectations intensify, there is strong strategic value in redesigning this workflow using responsible AI to capture meaningful efficiency gains.
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My vision centers on building a **human-AI collaboration model** that reduces manual workload, streamlines report preparation, and reallocates human effort toward higher-judgment review and oversight. Through a combination of supervised machine learning and generative AI, we can consolidate data from multiple sources, automatically identify relevant risk indicators, and generate structured, policy-aligned narratives. AI models equipped with few-shot and zero-shot learning can adapt to new case patterns without extensive training, enabling them to provide consistent and scalable support across case types. Analysts then shift their role from manual compilers to expert validators—reviewing AI-generated insights, refining narratives, and ensuring accuracy.

This approach aligns directly with **Porter’s Cost Leadership strategy**, which focuses on delivering products or services at the lowest sustainable cost while maintaining acceptable quality. AI becomes an engine for operational efficiency across several dimensions:

1. **Reduction in Manual Labor Costs**
Automating data gathering, anomaly detection, and narrative drafting reduces the amount of analyst time required per case. Even modest reductions in average handling time translate into substantial savings when applied across thousands of investigations annually.
2. **Minimization of Rework and Errors**
Manual processes introduce human errors such as missed data fields, inconsistent filters, or incomplete narratives. AI-driven validation and standardized narrative generation decrease error rates, lowering the cost of rework, secondary reviews, and post-audit remediation.
3. **Faster Turnaround Times**
Automated pre-analysis and narrative drafting accelerate the review cycle, allowing teams to complete more cases with the same staffing levels. This throughput improvement supports cost leadership by reducing dependency on overtime, temporary staffing, or additional headcount during peak periods.
4. **Scalability Without Proportional Cost Increases**
As case volumes increase due to regulatory change or economic shifts, AI tools scale seamlessly. Unlike human staffing models—where cost rises linearly—AI allows the organization to absorb additional volume with only marginal cost increases.
5. **Resource Reallocation to Higher-Value Work**
Time saved through automation can be reinvested into

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Current state	Describe the current state of your chosen organization. (Max. 500 words.) Start writing here:Our organization is widely recognized as a long-standing leader in the financial-services industry. We have a strong history of adopting advanced technologies across analytics, fraud detection, customer experience, and risk management. Today, innovative tools such as robotic process automation (RPA), supervised machine learning models, and digital workflow platforms are used to streamline operations, reduce errors, and enhance regulatory compliance. These technologies automate manual data entry, accelerate quality checks, and support risk scoring—allowing teams to handle high volumes of work more efficiently. A major recent advancement is the development of our own in-house large language model (LLM). While powerful, this model is still in the early stages of integration and has not yet been embedded into frontline operational workflows. Its potential remains underutilized, representing a significant opportunity for transformation. Organizational leadership actively encourages experimentation with AI, and many teams are exploring pilots to modernize manual processes, improve decision quality, and elevate customer and regulator-facing outputs. The current challenge—and opportunity—is to scale these technologies across business functions, especially in high-impact areas such as AML investigations, customer servicing, data validation, and internal reporting. Our organization’s competitive strategy aligns most closely with Cost Leadership, supported by early investment in innovative tools that make our operations more efficient than those of competitors. As a large financial institution managing millions of transactions and regulatory obligations, reducing operational costs while improving quality is central to our strategic advantage. Integrating AI—including supervised learning systems and the proprietary LLM—directly

executive.mit.edu	enables this strategy by minimizing manual workload, reducing rework, and shortening turnaround times. Analysts can shift away from repetitive data gathering toward higher-value review, oversight, and decision-making. Cost leadership is further reinforced by scalable AI infrastructure. Because the in-house LLM is a proprietary asset, once integrated, it offers a low marginal cost for each new use case. This allows the organization to absorb additional volume without proportional increases in staffing or operational spend. Faster processing, fewer errors, and consistent output quality position the organization to operate at lower cost relative to peers who rely on more traditional, labor-intensive workflows. While cost leadership is the primary strategic focus, the adoption of advanced analytics and proprietary AI capabilities also contributes to subtle differentiation. The ability to deliver faster, more accurate, and more consistent compliance outputs strengthens our reputation with regulators and internal stakeholders—enhancing trust and reinforcing market leadership. Overall, the combination of innovative technology deployment and a clear cost-leadership strategy positions the organization to continue leading the industry. Scaling AI integration—especially the in-house LLM—will accelerate efficiency gains, reduce operational costs, and strengthen our competitive edge in an increasingly automated and data-driven financial landscape.
Proposed initiative	Outline your proposed initiative for using AI in your chosen organization to achieve competitive advantage. (Max. 800 words.) Start writing here: Our organization has long been recognized as a pioneer in fintech and financial services, consistently deploying advanced technologies to modernize operations and maintain regulatory excellence. Although we have developed our own in-house large language model (LLM), this asset has not yet been fully integrated into everyday workflows. With leadership encouraging experimentation and broader adoption of AI, we have a strategic opportunity to leverage automation to redesign high-volume processes, elevate decision quality, and strengthen our position as an industry leader. The initiative I am proposing uses generative AI, supervised learning, and automated validation to transform analytics and investigative processes—directly advancing the organization’s competitive strategy of cost leadership.

The proposed solution centers around three AI capabilities that will be embedded across operational workflows. Our in-house LLM will be used to generate structured, policy-aligned investigative narratives and to summarize large amounts of information from multiple internal systems. Supervised learning models will validate dashboards, reports, and case outputs by detecting missing data, incorrect filters, inconsistent logic, and anomalies prior to distribution. AI-based data consolidation and evidence ranking will integrate key information from the numerous systems analysts currently use, highlighting the most relevant insights for decision-making. These capabilities will shift analysts from spending the majority of their time on manual data collection toward reviewing, refining, and applying judgment to AI-generated outputs.

A wide range of high-volume processes will benefit from this integration. AI will consolidate data from systems used for AML, KYC, and customer risk reviews, reducing the hours analysts spend switching between platforms. Automated quality checks will verify that dashboards and reports meet defined business rules. Narrative generation will standardize and accelerate the production of case summaries and investigation write-ups. Anomaly detection will highlight inconsistencies, gaps, or emerging risks. AI will also suggest next steps based on historical patterns and analyst behavior, helping guide deeper reviews where needed. Even triage activities—determining which cases require human review—can be partially automated. Together, these capabilities reduce manual workload and increase the speed, accuracy, and consistency of analytical output.

The integration of AI strongly supports the organization's cost-leadership strategy. Automating data gathering, validation, and narrative drafting reduces the labor required per report or investigation. Fewer errors and less rework lower downstream costs associated with remediation and audit findings. With AI handling much of the preliminary work, the same number of analysts can manage significantly higher volumes without proportional increases in staffing. Leveraging our proprietary LLM also reduces long-term technology expenses because it eliminates the need for costly external AI platforms. Taken together, these efficiencies allow the organization to operate at a lower cost base than peers who continue to rely primarily on manual processes. Although the initiative will also create differentiation benefits—such as improved narrative quality and stronger regulatory confidence—the primary advantage lies in delivering greater output at a significantly lower operational cost.

Rolling out this initiative requires a strong technical foundation. The in-house LLM must be integrated with case-management systems, data warehouses, and workflow tools. Supervised models need

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access to a curated feature store and labeled datasets built from historical defects and incident logs. Mature MLOps capabilities are essential for continuous monitoring, model retraining, version control, and drift detection. Explainability components are needed to meet audit requirements by showing which data points influenced a model's recommendations. Security measures such as encryption, access controls, and adherence to data-governance standards ensure the responsible use of sensitive customer information. These technical requirements are essential for delivering a solution that is stable, scalable, and compliant.

Effective managerial leadership is equally critical. As the technical lead, I guide the implementation from a technology perspective and ensure alignment between business needs and technical design. I translate requirements into detailed specifications, oversee system architecture, select appropriate tools, and ensure seamless integration across systems. I establish coding standards, define best practices, and manage quality assurance to maintain high delivery standards. I also identify technical risks early, troubleshoot issues, and provide mentorship to developers to strengthen our team's capabilities. Beyond my role, the initiative requires engagement from stakeholders who validate business requirements, analysts who test and refine AI outputs, Risk and Controls teams who ensure compliance, and Project Managers who coordinate activities and facilitate agile delivery. Change-management partners and Business Process Owners support training, adoption, governance, and long-term operational alignment.

The initiative is deeply aligned with the organization's IT strategy, which emphasizes automation, modernization, and the expansion of AI-driven capabilities. By leveraging our proprietary LLM and integrating it into workstreams that remain highly manual, we reinforce our commitment to innovation and operational resilience. This initiative supports the IT organization's mission to reduce friction, strengthen compliance, and deliver scalable technology that positions the company for future growth.

Customers will benefit from faster processing times, fewer delays, and more accurate reviews, as inconsistencies and data gaps are caught automatically. Suppliers and external partners will benefit from clearer communication, improved data accuracy, and greater predictability in the processes that rely on our outputs. The overall impact is tighter quality control, improved trust, and a more reliable operating environment for all parties involved.

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**Plan of action and
criteria for success**

Detail your plan of action for using AI in your chosen organization.
(Max. 800 words.)

Start writing here: The following initiative outlines how our organization can deploy AI to streamline analytical workflows, improve quality, and support our broader strategy of cost leadership while preparing for the future of automation.

Project Plan Overview and Timeline

The project begins with a discovery and planning phase focused on understanding current workflows, identifying data sources, and defining business rules. Over the first three weeks, we will map existing processes and gather labeled examples to support supervised learning. The next phase, lasting approximately two to six weeks, will focus on designing the system architecture, including integration paths for our in-house LLM, data pipelines, and downstream applications.

From weeks seven to twelve, we will train the supervised learning models, configure narrative-generation prompts, and run initial quality tests. A controlled pilot will follow, during which analysts will validate AI outputs and provide iterative feedback. This phase will last four to six weeks and will serve as the foundation for broader deployment. The final stage involves scaling the system across additional teams, implementing continuous learning mechanisms, and establishing long-term monitoring and governance. Full rollout is expected between five and seven months from project initiation.

Skills Needed to Build or Acquire

Successful deployment requires a blend of analytical, technical, and operational skill sets. Machine learning engineers will develop supervised models, manage training pipelines, and perform drift monitoring. Natural language processing specialists or LLM engineers will design prompts and ensure grounded narrative generation. Data engineers will build integrations and pipelines required to consolidate information from numerous systems.

MLOps experts are essential for deploying and maintaining models in production. Analysts with deep business knowledge will support labeling, evaluate results, and ensure that AI outputs align with policies and regulatory expectations. A strong change-management

function is also necessary to support user training and adoption. While most of these skills exist internally, contracting or temporarily augmenting the team may accelerate delivery.

Key Stakeholders

The initiative impacts multiple groups and requires alignment across stakeholders. Business stakeholders validate requirements and ensure the solution meets operational needs. Analysts play a central role in testing AI outputs and helping refine model behavior. Risk and Controls teams ensure that all AI components comply with governance standards, regulatory guidelines, and model-risk requirements. Technology teams handle development, integration, infrastructure, and security.

A dedicated Project Manager coordinates timelines, removes blockers, and facilitates communication across teams. Business Process Owners provide oversight on how workflows will evolve, while Change Management ensures that users are trained and organizational readiness is addressed early.

Organizational Aspects Involved

Rolling out AI requires careful consideration of how workflows and responsibilities shift. Analysts will transition away from manual data gathering and validation toward reviewing AI-generated insights and conducting complex investigations. This shift demands training, updated SOPs, and clear communication to avoid resistance. Governance structures must be updated to include model approval checkpoints, audit trails, and continuous monitoring.

Additionally, leaders must foster a culture that views AI as a partner rather than a threat. This means highlighting success stories, creating feedback mechanisms, and celebrating efficiency gains rather than framing automation strictly as cost reduction.

Managing the Initiative

As the technical lead, I will manage the initiative by aligning business and technical teams, ensuring clarity around requirements, and guiding the architectural design. I will enforce coding standards,

establish best practices, and lead quality assurance efforts. I also take responsibility for risk identification, troubleshooting, and mentorship for team members. The project will follow agile delivery, with iterative releases, frequent demos, and structured checkpoints to ensure transparency and course correction.

Special Considerations

Several special considerations must be addressed. Regulatory constraints require all AI-generated output to be explainable and grounded in the underlying data. Data privacy must be treated with the highest priority, requiring strict access controls and encryption. Change management is another critical factor; without proper communication and user engagement, adoption may lag even if the technology is sound. Ensuring that the AI complements rather than replaces human expertise is key to gaining user trust.

Ethical Concerns and Mitigations

AI systems can introduce bias if trained on inconsistent or incomplete data. To mitigate this risk, training data will be reviewed across teams and refreshed regularly. Overreliance is another ethical concern—analysts may defer too heavily to AI recommendations. Human-in-the-loop review will remain mandatory, and AI outputs will include evidence links to promote critical thinking. Transparency, auditability, and strict adherence to internal model-risk guidelines further ensure ethical deployment.

Impact on Jobs

The initiative is expected to shift roles rather than eliminate them. Manual tasks such as data extraction and routine validation will decrease, allowing analysts to focus on high-value work such as judgment-based reviews and investigations. New roles will emerge in model stewardship, automation oversight, and quality governance. The organization will invest in training and upskilling employees, enabling them to adapt to more technical and analytical responsibilities. This approach aligns with the idea of addressing human scarcity rather than work scarcity—AI extends capabilities rather than reduces need for talent.

Criteria for Success

Success will be measured through reduction in manual effort, lower error rates, faster turnaround times, and stronger compliance outcomes. Additional metrics include user adoption rates, analyst satisfaction, model accuracy, and reduction in post-distribution rework. Meeting these criteria will demonstrate both operational value and strategic alignment.

Extending the Roadmap

After the initial deployment, the roadmap includes expanding AI to other business lines, integrating RPA for automated actions, enhancing predictive models, and building a unified AI platform for analytics, validation, and investigations. Long term, this capability positions the organization to remain a cost leader while shaping the future of AI-enabled financial operations.

3. Rubric

	Criteria not met	Criteria met	Good	Exceptional
Executive summary	No submission. OR The response does not provide an executive summary that includes key points regarding the proposed initiative, information about the chosen organization, or any indication of how the	The response attempts to provide an executive summary, some key points regarding the proposed initiative, information about the chosen organization, and some indication of how the organization can benefit from the use	The response provides a well-structured executive summary, outlining key points regarding the proposed initiative, and information about the chosen organization, as well as insight into how the organization can benefit from the use	The response provides a comprehensive, well-structured executive summary, which effectively summarizes key points regarding the proposed initiative, as well as relevant information about the chosen organization and how it can benefit

	Criteria not met	Criteria met	Good	Exceptional
	organization can benefit from the use of AI in terms of Porter's strategies.	of AI in terms of Porter's strategies.	of AI in terms of Porter's strategies.	from the use of AI in terms of Porter's strategies.
Current state	No submission. The response does not provide insight into the organization's current use of technology and its position in relation to competitors.	The response attempts to provide insight into the organization's current use of technology and its position in relation to competitors.	The response provides adequate insight into the state of the organization's current use of technology and its position in relation to competitors.	The response provides a detailed, coherent, and insightful description of the current state of the organization's use of technology and its position in relation to competitors.
Proposed initiative	No submission. OR The response does not describe how AI could be used in the organization.	The response attempts to describe how AI could be used in the organization.	The response provides an adequate description of how AI could be used in the organization. The submission shows insight into the benefits and challenges of the use of AI and reflects on business or IT strategy, discusses technical and leadership requirements, and highlights the impact on competitive advantage.	The response provides a detailed, coherent, and insightful description of how AI could be used in the organization. The submission shows keen insight into the benefits and challenges of the use of AI, reflects on business or IT requirements, and highlights the impact on competitive advantage.
Plan of action and criteria for success	No submission. OR	The response attempts to describe a plan of action and	The response describes a realistic plan of action and	The response describes a detailed and sound plan of action and the

	Criteria not met	Criteria met	Good	Exceptional
	The response does not provide a description of the plan of action or the success criteria for implementing AI.	the success criteria for implementing AI.	the success criteria for implementing AI.	success criteria for implementing AI.
Structure and logic of writing: <i>Clear and logically structured submission that adheres to the word limit</i>	No submission. OR The response is unstructured, unclear, and lacks succinctness.	The response is logically structured, clear enough to comprehend, and mostly succinct.	The response is well structured in terms of logic, clarity, and succinctness.	The response is exceptionally well structured in terms of logic, clarity, and succinctness.